

Problem 16.1

$$1(n_1 - n_2) + 1(n_2 - n_3) - 1(n_1 - n_3) = 1$$

$$0 = 1$$

Problem 16.2

a) They ~~all~~ have to be orthogonal to the other two.
Two of them.

$$\dim N(A) + \dim C(A^T) = n$$

b) and $\dim N(A^T) + \dim C(A) = m$

$$A = CR^T$$